

**POLIMI GRADUATE
SCHOOL OF MANAGEMENT**

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

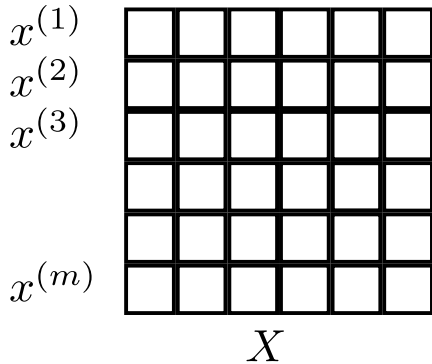
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- ▶ Introduction
- ▶ Data Preparation
- ▶ Sentiment Analysis
- ▶ Topic Modeling

NATURAL LANGUAGE PROCESSING

- From structured data to unstructured data



$x^{(1)}$: Walking across the sitting room, I turn the television off.

$x^{(2)}$: @UnitedAirlines second flight in a month #cancelled thanks a lot #worstairlineever

$x^{(3)}$: $E = mc^2$

```
import pandas as pd
```

$x^{(4)}$:

```
df = pd.read_csv("data.csv", index_col=0)
print(df.shape)
```

SOME IMPORTANT QUESTIONS

- ▶ We would like to exploit “all” that we know on structured data
- ▶ How can we represent text in a structured way?
- ▶ Which are the limits/problems of a representation?

THE ELEMENTARY UNIT

The elementary unit of a digital image is the pixel, what about text?

→ words? letters?

- ▶ We need a procedure to split a document into tokens → **tokenization**
- ▶ The same words can have different meanings
- ▶ The same meaning can be expressed by different words

CONTEXT AND SEQUENCES

- ▶ The meaning of a word depends on its context
- ▶ Context is not always defined by proximity

This summer I went to Italy. I had a lovely time there, engaging in both recreational and cultural <b1>, meeting old friends and making new ones [...] but most of all I had my food, <b2>.

CONTEXT AND SEQUENCES

- ▶ Order plays an important role

Only he can play that instrument

He only can play that instrument

- ▶ Meaning might be hard to understand without context (e.g., sarcasm)

I am having a lovely time

I am having a *lovely* time

TEXT CLEANING

- ▶ Convert to lower case
- ▶ Remove punctuation
- ▶ Remove numerical values
- ▶ Typos
- ▶ Remove special characters ([?@)
- ▶ Remove stop words (the, it, etc)
- ▶ Remove special description words ([chorus], [fade], [applause])

TEXT CLEANING

- ▶ Tokenize text
 - O'Neill → [o] [neill], [o'neill]?
 - aren't → [arent], [are][nt]?
- ▶ Create bi-grams or tri-grams ([United Kingdom] vs [United][Kingdom])
- ▶ Normalization:
 - Stemming (car, cars, car's, cars' → car;)
 - Lemmatization (am, are, is → be)

TEXT REPRESENTATION - BAG OF WORDS

- ▶ Corpus: a collection of text
- ▶ Document-Term Matrix: word counts in matrix format
- ▶ Bag of Words
 1. **Only he** can play that instrument
 2. **He only** can play that instrument

	he	only	can	play	that	instrument
D1	1	1	1	1	1	1
D2	1	1	1	1	1	1

TEXT REPRESENTATION - TF-IDF

- ▶ TF-IDF: Term Frequency - Inverse Document Frequency

$$\text{TF-IDF} = f_{t,d} \times \text{idf}(t, D)$$

where

- $f_{t,d}$: the number of times that term t occurs in document d .
- $\text{idf}(t, D) = \log \frac{|D|}{|\{d \in D: t \in d\}| + 1}$: log of the inverse of the number of documents containing term t .

SENTIMENT ANALYSIS

[Un]supervised task of extracting a sentiment score from text

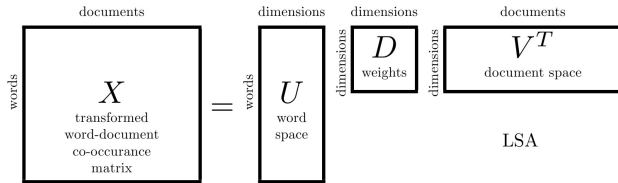
- ▶ Sentiment Labels: Each word is labeled in terms of
 - Polarity: negative (-1) or positive (+1)
 - Subjectivity: subjective (0) or fact (+1) [TextBlob]
- ▶ Sentiment of words can vary based on where it is in a sentence.
 - Negation multiplies the polarity by -0.5
- ▶ The challenge? Irony and sarcasm
- ▶ What practical application?

TOPIC MODELING

The task of discovering hidden themes (topics) in the documents of a corpus

- ▶ Word co-occurrence patterns are used to cluster documents into thematic groups
 - “screen”, “battery”, and “charging” → Smartphone hardware
- ▶ “Cluster” ownership is not enough, we need to extract the topic
- ▶ Unsupervised: number of topics and topic labels are unknown
- ▶ What practical application?

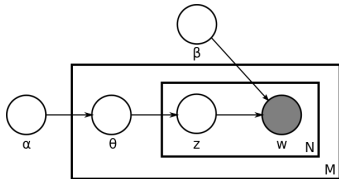
TOPIC MODELING - LSA



Latent Semantic Analysis (LSA)

- **Singular Value Decomposition (SVD)** of the Document-Term Matrix

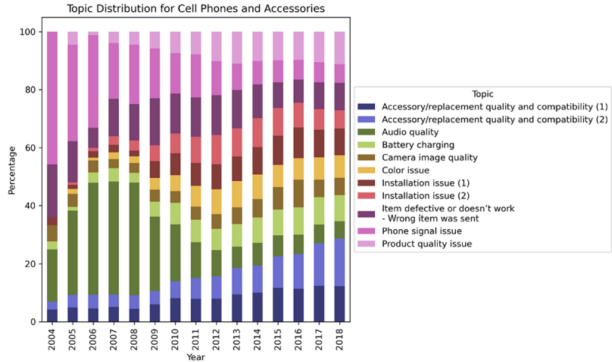
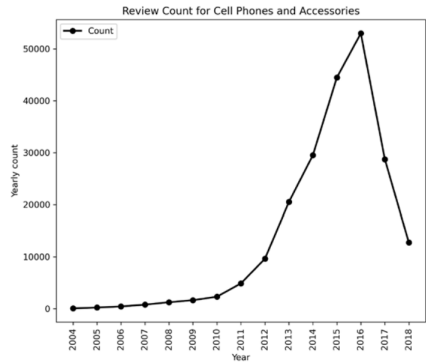
TOPIC MODELING - LDA



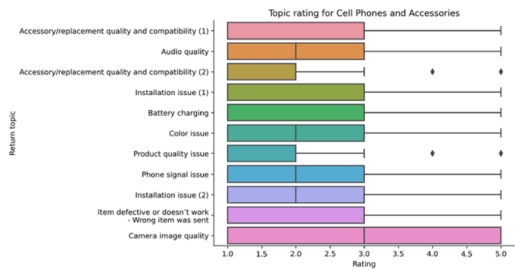
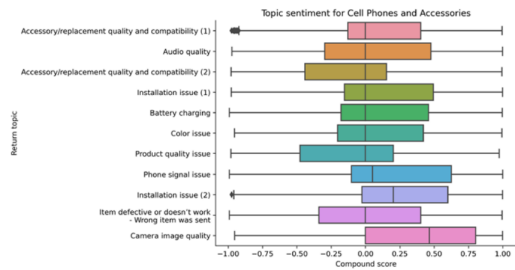
Latent Dirichlet Allocation (LDA)

- ▶ **Documents (M) are probabilistic distribution over topics:** let say that a document is $p_i\%$ of topic i .
 - e.g., $\theta_j = \{\text{Sport} : 0.5, \text{Politics} : 0.3, \text{Finance} : 0.2\}$
- ▶ **Topics are probabilistic distribution over words:** given a topic chosen according to the distribution of the document, we generate a word according to the topic distribution
 - e.g., $\phi_{Sport} = \{\text{Ball} : 0.5, \text{Goal} : 0.3, \text{Match} : 0.2\}$

METADATA: TOPICS IN TIME



TOPIC + SENTIMENT AND METADATA



THANK YOU